Experimental Data Collection Guide

Introduction to Experimental Research

Experimental research is the gold standard for establishing cause-and-effect relationships. This guide will help you design, conduct, and document experiments effectively.

WHAT IS AN EXPERIMENT?

An experiment is a research method where the researcher **manipulates** one or more **independent variables** and measures their effect on **dependent variables** while **controlling** for **extraneous variables**.

Key Characteristics

✓ **Manipulation**: Researcher actively changes conditions
✓ **Control**: Extraneous factors are held constant
✓ **Randomization**: Participants randomly assigned to conditions
✓ **Replication**: Can be repeated by other researchers

TYPES OF EXPERIMENTAL DESIGNS

1. True Experimental Design

Between-Subjects Design (Independent Groups)

Different participants in each condition.

Example:

- Group A: Receives new teaching method
- Group B: Receives traditional teaching method
- Compare outcomes between groups

Advantages:

- No practice effects
- Simpler analysis

Disadvantages:

- Needs more participants
- Individual differences may confound results

Within-Subjects Design (Repeated Measures)

Same participants experience all conditions.

Example:

- All students try Method A first
- Then all students try Method B
- Compare each student's performance across methods

Advantages:

- Fewer participants needed
- Controls for individual differences

Disadvantages:

- Order effects (practice, fatigue)
- Time-consuming

Mixed Design

Combination of between and within-subjects factors.

Example:

- Factor 1 (between): Male vs. Female students
- Factor 2 (within): Pre-test vs. Post-test scores

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2. Quasi-Experimental Design

Used when random assignment is not possible.

Examples:

- **Non-equivalent control group**: Use existing classes as experimental and control groups
- **Time series**: Measure repeatedly before and after intervention
- **Natural experiments**: Study naturally occurring variations

When to Use:

- Ethical constraints prevent randomization
- Practical limitations (existing groups)
- Field settings where control is limited

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3. Pre-Experimental Design

Least rigorous; used for exploratory studies.

Types:

- **One-shot case study**: Single group, treatment, then measure
- **One-group pretest-posttest**: Measure before and after treatment
- **Static group comparison**: Two groups compared post-treatment only

⚠️ **Limited internal validity -** cannot confidently establish causation

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VARIABLES IN EXPERIMENTS

Independent Variable (IV)

What you **manipulate** or change.

Example: Teaching method (traditional vs. technology-based)

Levels: The different conditions or values of the IV

- Level 1: Traditional method
- Level 2: Technology-based method

Dependent Variable (DV)

What you **measure** as the outcome.

Example: Student exam scores

Operationalization: How specifically you'll measure it

- Final exam score (0-100 scale)
- Measured one week after instruction ends

Extraneous Variables

Factors that could affect the DV but are not of primary interest.

Examples:

- Student's prior knowledge
- Time of day of testing
- Classroom temperature
- Teacher enthusiasm

Confounding Variables

Extraneous variables that **systematically vary** with the IV, making it impossible to determine the true cause of effects.

Example: If the technology group also has a more experienced teacher, you can't tell if differences are due to method or teacher.

EXPERIMENTAL CONTROL TECHNIQUES

1. Randomization

Random Assignment: Participants randomly allocated to conditions

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Participants → [Random Assignment] → Group A (Treatment)
                                   → Group B (Control)
```

Benefits:

- Equalizes groups on known and unknown variables
- Strengthens internal validity

2. Matching

Pair participants on key variables before assigning to groups.

Example:

- Pair students with similar GPA
- Assign one from each pair to each group

3. Counterbalancing

Vary the order of conditions to control order effects.

Complete Counterbalancing:

- Half experience: A then B
- Half experience: B then A

Latin Square: For multiple conditions (A, B, C, D):

```
Participant 1: A → B → C → D
Participant 2: B → C → D → A
Participant 3: C → D → A → B
Participant 4: D → A → B → C
```

4. Standardization

Keep procedures identical across all conditions.

Standardize:

- Instructions (use scripts)
- Environment (same room, time)
- Materials (identical quality)
- Duration (same length)
- Measurement (same tools, raters)

5. Control Groups

Types:

- **No-treatment control**: Receives nothing
- **Placebo control**: Receives inactive treatment
- **Active control**: Receives standard/alternative treatment

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STEP-BY-STEP EXPERIMENTAL PROCEDURE

Phase 1: Planning (Before Data Collection)

Step 1: Define Your Hypothesis

H_0 (Null): There is no difference in exam scores between traditional and technology-based teaching methods.

H_1 (Alternative): Students taught with technology-based methods will score significantly higher on exams than those taught with traditional methods.

Step 2: Identify Variables

- **IV:** Teaching method (2 levels)
- **DV:** Exam score (0-100)
- **Controlled variables:** Teacher, content, duration, classroom
- **Potential confounds to avoid:** Prior knowledge, motivation

Step 3: Select Participants

Sample Size: Calculate using power analysis

- Effect size estimate
- Desired power (typically 0.80)
- Significance level (typically $\alpha = 0.05$)

Inclusion Criteria:

- Currently enrolled students
- No prior exposure to material
- Consent provided

Exclusion Criteria:

- Previous course failures
- Irregular attendance
- Special accommodations that would affect outcome

Step 4: Design Procedures

Timeline:

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Week 1: Recruit and randomly assign participants
Week 2: Administer pre-test
Weeks 3-6: Deliver interventions
Week 7: Administer post-test
Week 8: Debrief and analyze data
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Materials Needed:

- Instructional materials
- Pre/post-test forms
- Consent forms
- Data recording sheets
- Equipment (projector, tablets, etc.)

Phase 2: Pilot Testing

Purpose: Identify and fix problems before main study

What to Test:

- [ ] Instructions are clear
- [ ] Procedures are feasible
- [ ] Timing is appropriate
- [ ] Measures work correctly
- [ ] No unexpected confounds

Pilot Sample: 5-10 participants not in main study

Revise Based On:

- Participant feedback
- Procedural challenges
- Measurement issues
- Time constraints

Phase 3: Data Collection

Session Preparation Checklist

- [ ] Room is set up identically for all sessions
- [ ] All materials are ready and organized
- [ ] Equipment is tested and working
- [ ] Data sheets are prepared
- [ ] Consent forms are available
- [ ] Random assignment list is ready

During Each Session

1. Welcome and Consent (5 minutes)

> Script: "Thank you for participating. Before we begin, please read and sign this consent form. Your participation is voluntary and all data will be kept confidential."

2. Administer Pre-test (if applicable)

- Use standardized instructions
- Time consistently
- No feedback during test

3. Deliver Treatment

- Follow protocol exactly
- Take notes on any deviations
- Maintain neutral demeanor

4. Administer Post-test

- Same conditions as pre-test
- No hints or assistance
- Collect immediately

5. Debrief

- Explain study purpose

- Answer questions
- Provide contact information

Data Recording

Immediate Recording: Record data immediately after each session to prevent memory errors.

Data Sheet Format:

Participant ID: _____
Condition: _____
Date: _____
Time: _____
Pre-test Score: _____
Post-test Score: _____
Notes/Observations: _____

Digital Backup:

- Enter data into spreadsheet daily
- Keep backup copies
- Check for entry errors

Phase 4: Quality Control

During Study

Monitor for:

- [ ] Procedural drift (procedures changing over time)
- [ ] Participant attrition (drop-outs)
- [ ] Equipment malfunctions
- [ ] Environmental changes
- [ ] Contamination between groups

Maintain Blinding When Possible:

- **Single-blind**: Participants don't know which condition they're in
- **Double-blind**: Neither participants nor researcher knows (uses coded materials)

After Each Session

Check:

- [ ] All data recorded completely
- [ ] Consent forms signed
- [ ] Materials returned/cleaned
- [ ] Notes documented
- [ ] Any incidents logged

ETHICAL CONSIDERATIONS

Informed Consent

Participants must understand:

- Study purpose (general)
- Procedures
- Risks and benefits
- Right to withdraw
- Confidentiality protections

Deception

If necessary for valid results:

- Must be justified
- Cannot involve significant risk
- Full debriefing required afterward

Debriefing

After participation, explain:

- True purpose of study
- Why deception (if any) was necessary
- How data will be used
- Opportunity for questions

Confidentiality

- Use participant IDs, not names
- Store consent forms separately from data
- Limit access to identifying information
- Report only aggregate results

DATA MANAGEMENT

Organization System

File Structure:

```
Experiment_Teaching_Methods_2025/
├── 01_Planning/
│   ├── protocol.pdf
│   ├── ethics_approval.pdf
│   └── materials_list.xlsx
├── 02_Consent_Forms/
│   └── signed_consent_forms/
├── 03_Raw_Data/
│   ├── pretest_scores.xlsx
│   ├── posttest_scores.xlsx
│   └── session_notes.docx
├── 04_Analysis/
│   ├── cleaned_data.xlsx
│   ├── spss_output.spv
│   └── analysis_script.R
└── 05_Results/
    ├── figures/
    ├── tables/
    └── manuscript_draft.docx
```

Data Cleaning Checklist

- [ ] Check for missing data
- [ ] Verify data entry accuracy (double-check 10% of entries)
- [ ] Identify and handle outliers
- [ ] Code categorical variables consistently
- [ ] Transform variables if needed

- [ ] Calculate composite scores
- [ ] Check assumptions for statistical tests

COMMON THREATS TO VALIDITY

Internal Validity (Can you conclude causation?)

History: External events occur during study

- *Control:* Use control group experiencing same timeframe

Maturation: Participants change naturally over time

- *Control:* Use control group, keep study brief

Testing: Pre-test affects post-test performance

- *Control:* Use control group, or eliminate pre-test

Instrumentation: Measurement changes over time

- *Control:* Calibrate equipment, train raters

Selection: Groups differ before treatment

- *Control:* Random assignment

Attrition: Differential drop-out rates

- *Control:* Track drop-outs, analyze their characteristics

External Validity (Can you generalize?)

Population Validity: Sample represents target population?

- *Improve:* Diverse recruitment, representative sampling

Ecological Validity: Lab setting represents real world?

- *Improve:* Field experiments, realistic tasks

Temporal Validity: Findings hold across time?

- *Improve:* Replicate at different times

REPORTING YOUR EXPERIMENT

Method Section Structure

Participants

> "Eighty undergraduate students (40 male, 40 female; Mage = 21.3 years, SD = 2.1) from the University of Buea participated in exchange for course credit. Participants were randomly assigned to technology-based (n = 40) or traditional (n = 40) instruction conditions."

Materials

Describe all materials used:

- Tests/questionnaires (include sample items)
- Equipment (model numbers)
- Stimuli (with examples or images)
- Software (version numbers)

Procedure

Chronological, detailed description:

- What participants did
- What experimenter did
- Exact timing
- Any randomization or counterbalancing

Design

State the experimental design:

> "The study employed a 2 (teaching method: traditional vs. technology) × 2 (time